

ai-os-new

State Inconsistency Re-Audit Report

Verification audit after bug fixes. Original: 14 findings. Remaining: 14 findings.

Severity	Count
CRITICAL	3
HIGH	5
MEDIUM	5
LOW	1
Total	14

SI-01

CRITICAL

Key status tracked in 7+ incompatible state machines

Files: key-health.ts, key-state-store.ts, key-lifecycle.ts, key-analytics.ts, health-service.ts, health-score-service.ts, key-state-projection.ts

Problem: Seven independent subsystems track key status with different enums and transition logic. None is authoritative. Can diverge simultaneously.

Fix: Designate KeyStateStore as single source of truth. Add reconciliation method. All subsystems update through KeyStateStore.

SI-02

CRITICAL

Cascade delete failure on key removal

Files: key-service.ts, key-state-store.ts, key-lifecycle.ts, key-health.ts, virtual-key-service.ts, session-affinity-store.ts

Problem: removeKey() only calls registry.removeKey() and emits KEY_REMOVED. Does NOT clean up KeyStateStore, KeyLifecycle, KeyHealth, VirtualKeyService, or SessionAffinityStore.

Fix: Add cascadeCleanup(keyId) method calling health.cleanupKey(), keyStateStore.remove(), virtualKeyService.revokeByRealKeyId(), sessionAffinityStore.evictByKey().

SI-03

CRITICAL

Config rollback doesn't clear ConfigService overlays or notify downstream

Files: config-history.ts, config-service.ts, settings-service.ts, router-config-manager.ts

Problem: rollback() mutates global CONFIG but overlays are NOT cleared. Next getKeys() re-applies stale overlays, partially undoing rollback. SettingsService and RouterConfigManager not notified.

Fix: Emit CONFIG_ROLLED_BACK event. ConfigService clears overlays. SettingsService and RouterConfigManager reinitialize.

SI-04

HIGH

Cross-tab sync has no freshness/vector-clock ordering

Files: cross-tab-state.ts

Problem: handleSyncResponse unconditionally overwrites local state. No timestamp comparison. Stale data from another tab can overwrite fresh data.

Fix: Add observedAt timestamp. Only overwrite if incoming is more recent than local.

SI-05

HIGH

Snapshot restore doesn't invalidate caches or projections

Files: snapshot-service.ts, cache-service.ts, key-state-store.ts, health-service.ts

Problem: restore() replaces kernel state but doesn't invalidate CacheService, KeyStateStore, HealthService results, or ProviderTracker.

Fix: Emit SNAPSHOT_RESTORED event. Have all caches/projections listen and reset.

SI-06

HIGH

HealthScoreService.computeScoreFromState() always returns null

Files: health-score-service.ts

Problem: Accesses providerTracker via unsafe type cast but it's never injected. getScore() returns null when cache expires. Service is non-functional.

Fix: Add setProviderTracker() method. Call during bootstrap. Or accept HealthScoreInput directly.

SI-07

HIGH

Event replay doesn't reset projections before applying

Files: replay-engine.ts, projection-registry.ts

Problem: ReplayEngine replays events but projections are NOT reset first. Replayed events applied on top of existing state = incorrect results.

Fix: Call registry.resetAll() before starting replay.

SI-08

HIGH

Virtual keys dangle after real key deletion

Files: virtual-key-service.ts, key-service.ts

Problem: No listener for KEY_REMOVED events. Virtual keys with deleted realKeyId remain active:true. resolve() returns them as valid.

Fix: Add KEY_REMOVED listener in VirtualKeyService.init() that revokes matching virtual keys.

SI-09

MEDIUM

SessionAffinityStore never reaps active bindings

Files: session-affinity-store.ts

Problem: reapExpired() only removes pending-eviction bindings. Active bindings persist indefinitely even after sessions end.

Fix: Add maxBindingAgeMs (24h). Listen for SESSION_ENDED event.

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MEDIUM

MessageIndexService byRequestId index corrupted on eviction

Files: message-index-service.ts

Problem: When oldest message evicted, byRequestId.delete(removed.requestId) removes entry even if assistant message with same requestId still exists.

Fix: Use Map<string, Set<IndexedMessage>> or reference counter.

SI -
11

MEDIUM

AgentService.deleteAgent doesn't clean health/versions

Files: agent-service.ts, agent-health-monitor.ts, agent-version-service.ts

Problem: deleteAgent() doesn't clean AgentHealthMonitor.records/healthCache, AgentVersionService.cache, or AgentService.stats.

Fix: Emit AGENT_DELETED event. Have health/versions/stats listen and clean up.

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MEDIUM

BudgetService and ProviderBudget track costs independently

Files: budget-service.ts, provider-budget.ts

Problem: BudgetService uses STREAM_END + costCalculator. ProviderBudget uses startSession/endSession/recordUsage. Different event sources = different numbers.

Fix: Unify on single cost recording path. Emit COST_RECORDED event both systems listen to.

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MEDIUM

KeyLifecycle per-key state never evicted on key removal

Files: key-lifecycle.ts

Problem: No removeKey(id) method. lifecycleStates, errorCounters, successCounters, rotationTimers never evicted for deleted keys. Memory leak over time.

Fix: Add removeKey(id) that clears all per-key maps. Call from KeyService.removeKey().

SI -
14

LOW

CacheDecorator modelCache not invalidated on key rotation/compromise

Files: cache-decorator.ts

Problem: modelCache uses 2-min TTL. On KEY_REMOVED/KEY_COMPROMISED, stale model list served for up to 2 minutes.

Fix: Listen for key change events and clear modelCache entry. Or reduce TTL to 30s.